

Follow the Job

IT 520 · Week 2 Friday participation activity

Name _____

Work Parts 1-6 in order. Use the matching closed answer help on the Canvas page whenever you need it. Check, visibly revise at least one line, save, and submit this worksheet or a clear photo.

The machine we already know

STORAGE → MEMORY → CPU → MEMORY → STORAGE / OUTPUT
stored loaded compute result saved

- **Storage** keeps the input and finished file.
- **Memory** holds active input and working results.
- **CPU** performs the processor work.
- **Output** is not ready until the result has been saved.

Wednesday supplied this timing for the nightly report: **8 hours data movement + 2 hours processor work = 10 hours whole job.** Those numbers do not show which read, write, wait, or calculation used the time.

Part 1 — rebuild Wednesday's job

Put the steps in order and place each on storage, memory, CPU, or output. A step may use two places. Label a move or calculation as **data movement** or **processor work**; leave a state-only step blank. Do not divide the supplied hours among steps.

Step	What happens	Order	Place(s)	Work type, if applicable
A	The completed report is saved; workers can open it.			
B	The CPU calculates report totals from active data.			
C	Input records are kept in a stored file.			
D	Input records are copied into working memory.			
E	The CPU's result is held in memory.			

Check 1: C is first.

Part 2 — follow one instruction through the CPU

The program counter points to an instruction that means **add 1 to the value at memory address 201**. Before the instruction begins, **memory[201] = 7**.

Write **fetch**, **decode**, **execute**, or **write-back** beside each action.

Action	Stage
The CPU copies the next instruction into the instruction register and advances the program counter.	
The CPU determines that the instruction says “add 1” and that address 201 is the destination.	
The ALU produces the result 8. Memory address 201 still contains 7.	
The CPU stores 8 at memory address 201.	

Which stage changes the destination value in memory?

Where does this repeated four-stage cycle fit inside the nightly-report path from Part 1?

Check 2: fetch comes first.

Part 3 — compare two ten-hour jobs

Wednesday compared two ten-hour jobs on the same machine. **Condition A is the nightly report. Condition B is a different job** with opposite stage shares: same machine, same total, different work inside it.

Condition	Data movement	Processor work	Whole job
A	8 hours	2 hours	10 hours
B	2 hours	8 hours	10 hours

For **each** condition, suppose a faster CPU halves processor-work time without changing data-movement time. Calculate the new whole-job time.

Condition	New processor-work time	New whole-job time
A		
B		

What do the two results show about when a faster CPU matters?

Part 4 — explain what the total does and does not prove

For Condition A, complete:

- The ten hours establishes _____.
- It does not establish that the CPU used ten hours because _____.
- A faster CPU cannot reduce the supplied eight hours of _____.

Could the ten-hour total alone tell whether loading or saving took longer? Why or why not?

Part 5 — transfer the model to a web request

The same machine model still helps, but input arrives over a network. Order the steps and place each on **network**, **memory**, **CPU**, **storage**, or **output/browser**. A step may use more than one place.

Step	What happens	Order	Place(s)
F	Browser displays the returned catalog item.		
G	Server receives the request into memory.		
H	Browser sends a request across the network.		
I	CPU runs the catalog-service instructions.		
J	Service reads the requested record from storage into memory.		
K	Server sends the result across the network.		

Check 3: H begins the request. A networked job adds crossings; it does not replace storage, memory, and CPU with a new machine model.

Part 6 — place one observation

Circle one crossing where you could place timestamps. Name exactly what those timestamps would and would not establish.

Finish

- ☐ Nightly-report steps are ordered and placed on the Week 1 machine.
 - ☐ The fetch → decode → execute → write-back example is complete.
 - ☐ The ten-hour claim is bounded, and the two jobs show when a faster CPU helps.
 - ☐ The web request reuses the same model.
 - ☐ One observation is placed and its claim is limited.
 - ☐ I checked the key, visibly revised at least one line, and saved my work.
- Submit through the location paired with this activity. If none is visible, retain the file or a photo and contact the instructor.